Applying Probability and Statistics to Evaluate a Machine Learning Model

1. Problem Statement and Dataset

The objective is to determine whether an improved classifier performs meaningfully better than a baseline classifier, or whether an observed difference could plausibly arise from variation in the finite test sample.

Dataset: the Breast Cancer Wisconsin Diagnostic dataset distributed with scikit-learn. It contains 569 observations and 30 numeric features computed from digitized images of breast mass fine-needle aspirates. The classification target distinguishes malignant and benign cases. The dataset is loaded directly through sklearn.datasets.load_breast_cancer, so no manual download is required.

Baseline model: DecisionTreeClassifier with random_state=42. Improved candidate: ExtraTreesClassifier with 300 trees and random_state=42. The ensemble is considered the improved candidate because combining randomized trees can reduce the instability of a single decision tree; its superiority is tested rather than assumed.

2. Sampling and Experimental Methodology

A stratified 75%/25% train-test split is created with random_state=42. Stratification keeps the class proportions approximately consistent across the partitions. Both models are fitted only on the training set and evaluated on the exact same held-out test rows (143 test observations). The test set is not used to fit either model.

2.1 Sampling and estimation

Test accuracy is the sample proportion of correctly classified observations. It estimates the model's classification accuracy on the population represented by the test sample. Because the estimate is based on a finite sample, it is subject to sampling variability.

2.2 Paired bootstrap confidence interval

For each test observation, define a paired difference as 1 if the improved model is correct and the baseline is wrong, −1 if the baseline is correct and the improved model is wrong, and 0 if both models are either correct or wrong. The mean of these differences equals the observed accuracy difference. The code resamples test-row indices with replacement 10,000 times, using the same sampled indices for both models, and takes the 2.5th and 97.5th percentiles to form a 95% percentile bootstrap confidence interval.

2.3 Hypothesis test

The exact two-sided McNemar test is used because the two models make predictions on the same examples. It considers only discordant pairs: b is the number of examples where the baseline is correct and the improved model is wrong; c is the number where the baseline is wrong and the improved model is correct. Under the null hypothesis of equal error probabilities on discordant pairs, the smaller count follows an exact Binomial(n=b+c, p=0.5) test. The significance level is $\alpha = 0.05$.

Hypotheses: H_0 — the two classifiers have equal error probability on paired test cases; H_1 — their error probabilities differ. The test is two-sided because the question is whether performance differs, not merely whether it changes in a preselected direction.

3. Observations, Decision, and Limitations

Measure	Baseline: Decision Tree	Improved: Extra Trees
Test accuracy	92.31%	96.50%
Correct predictions (out of 143)	132	138

Observed accuracy difference (improved − baseline): +4.20 percentage points. In this fixed run, there are 3 discordant cases where the baseline is correct and the improved model is wrong (b=3), and 9 cases where the baseline is wrong and the improved model is correct (c=9).

The paired 95% bootstrap confidence interval for the accuracy difference is approximately −0.70 to +9.09 percentage points. It includes zero. The exact two-sided McNemar p-value is approximately 0.1460, which is greater than α=0.05.

Decision: Extra Trees has higher observed test accuracy in this experiment, but the difference is not statistically significant at the 5% level. We fail to reject the null hypothesis; this does not prove the models are identical. The available test sample does not provide strong enough evidence to conclude that the improvement is reliable rather than due to sampling variation.

Output

```
PS C:\Users\MSIS\Downloads\APS\assignments\evaluate_ml> python evaluate_ml_models.py
======================================================================
PROBABILITY AND STATISTICS FOR ML MODEL EVALUATION
======================================================================
Dataset: .. _breast_cancer_dataset:
Total samples: 569; features: 30
Train samples: 426; held-out test samples: 143
Random state: 42; test size: 25%

Baseline (Decision Tree) accuracy: 0.9231 (92.31%)
Improved (Extra Trees) accuracy:  0.9650 (96.50%)
Observed accuracy difference:     +0.0420 (+4.20%)
95% paired bootstrap CI:           [-0.0070, +0.0909]

Exact McNemar test (paired predictions)
Baseline correct / improved wrong (b): 3
Baseline wrong / improved correct (c): 9
Two-sided p-value: 0.1460
Significance level alpha: 0.05

Confusion matrix — baseline (rows=true, columns=predicted):
[[49  4]
 [ 7 83]]

Confusion matrix — improved (rows=true, columns=predicted):
[[50  3]
 [ 2 88]]

Classification report — baseline:
              precision    recall  f1-score   support

   malignant       0.88      0.92      0.90        53
      benign       0.95      0.92      0.94        90

    accuracy                           0.92       143
   macro avg       0.91      0.92      0.92       143
weighted avg       0.92      0.92      0.92       143

Classification report — improved:
              precision    recall  f1-score   support

   malignant       0.96      0.94      0.95        53
      benign       0.97      0.98      0.97        90

    accuracy                           0.97       143
   macro avg       0.96      0.96      0.96       143
weighted avg       0.96      0.97      0.96       143
```

```
STATISTICAL DECISION
The improved model has higher observed accuracy, but the evidence
is not strong enough to establish a reliable improvement at 5%.
The bootstrap interval includes zero and/or McNemar's p-value
is not below alpha. Do not claim statistical significance.

Interpretation note: the test set is a sample, so its result has
sampling uncertainty. The bootstrap CI quantifies uncertainty in
the paired accuracy difference; McNemar's test compares discordant
paired predictions. Neither test guarantees performance on every
future population or dataset.
```

Limitations

- The test set contains only 143 examples, so uncertainty remains and small differences are difficult to establish.
- A single train-test split can produce split-specific results. Repeated stratified cross-validation or repeated holdout evaluation could provide a more complete picture, provided the final test set remains untouched for final assessment.
- The bootstrap interval is a percentile interval based on resampling the observed test cases; it cannot represent every source of uncertainty, such as changes in the future population or data collection process.
- Accuracy treats false positives and false negatives equally. In a medical screening context, sensitivity, specificity, calibration, and the costs of errors should also be considered.
- The improved model was chosen as a candidate for this demonstration. Repeatedly tuning models against the same test set would cause test-set leakage and make the reported result optimistic.

4. Where Probability and Statistics Are Used

- **Stratified random sampling:** The dataset is split into training and test samples while preserving approximate class proportions.
- **Statistical estimation:** Test accuracy is a sample-based estimate of generalization performance.
- **Bootstrap resampling:** Sampling test indices with replacement approximates the sampling distribution of the paired accuracy difference.
- **Confidence interval:** The 95% interval gives a range of plausible values for the accuracy difference under the bootstrap procedure; inclusion of zero indicates that no improvement is compatible with the interval.
- **Hypothesis testing:** The exact McNemar test evaluates whether the pattern of paired wins and losses is inconsistent with equal error probabilities.
- **p-value and significance level:** The p-value (about 0.146) is compared with $\alpha=0.05$. Since $p > \alpha$, the null hypothesis is not rejected.
- **Probability model:** McNemar's exact test uses a Binomial(n=b+c, p=0.5) distribution under the null for the discordant pairs.

Reproducibility

Run evaluate_ml_models.py with Python 3 and the packages numpy, scipy, and scikit-learn installed. The random seeds, model settings, split, bootstrap replicate count, and statistical test are specified in the script. The report's numerical values correspond to those settings.

Suggested command: pip install numpy scipy scikit-learn

Then run: python evaluate_ml_models.py

References

1. scikit-learn documentation, Breast Cancer Wisconsin dataset: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

2. scikit-learn documentation, ExtraTreesClassifier: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.ExtraTreesClassifier.html

3. Fagerland, M. W., Lydersen, S., & Laake, P. (2013). The McNemar test for binary matched-pairs data: mid-p and exact tests. Statistical Methods in Medical Research, 22(4), 346–361.